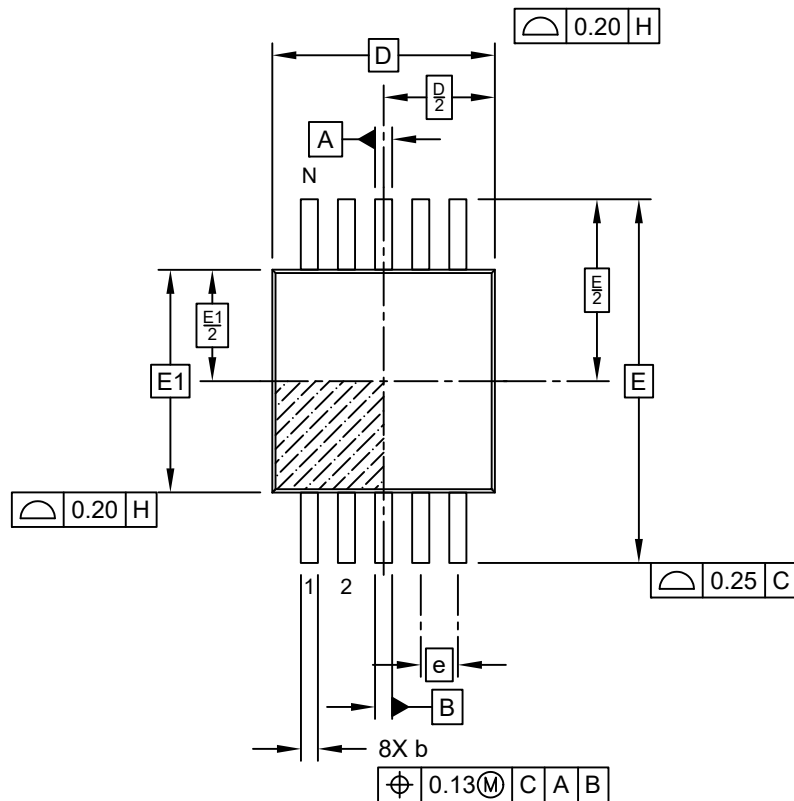
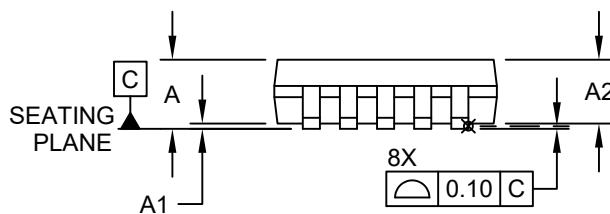


## 10-Lead Plastic Micro Small Outline Package (MS) - 3x3 mm Body [MSOP]

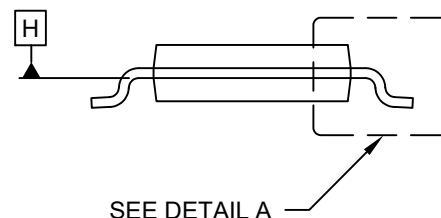
**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



TOP VIEW



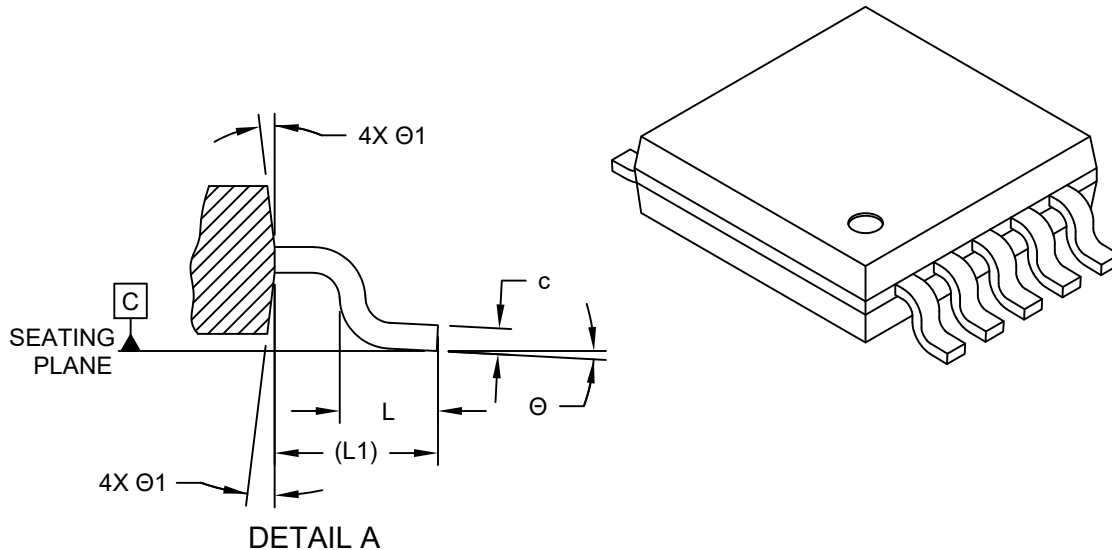
SIDE VIEW



END VIEW

## 10-Lead Plastic Micro Small Outline Package (MS) - 3x3 mm Body [MSOP]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



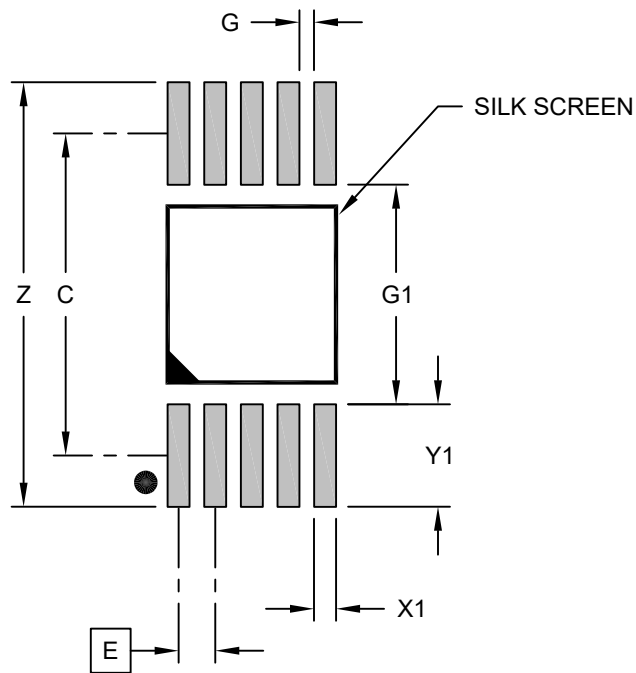
| Dimension Limits         | Units | MILLIMETERS |      |      |
|--------------------------|-------|-------------|------|------|
|                          |       | MIN         | NOM  | MAX  |
| Number of Pins           | N     | 10          |      |      |
| Pitch                    | e     | 0.50 BSC    |      |      |
| Overall Height           | A     | -           | -    | 1.10 |
| Molded Package Thickness | A2    | 0.75        | 0.85 | 0.95 |
| Standoff                 | A1    | 0.00        | -    | 0.15 |
| Overall Width            | E     | 4.90 BSC    |      |      |
| Molded Package Width     | E1    | 3.00 BSC    |      |      |
| Overall Length           | D     | 3.00 BSC    |      |      |
| Foot Length              | L     | 0.40        | 0.60 | 0.80 |
| Footprint                | L1    | 0.95 REF    |      |      |
| Foot Angle               | Ø     | 0°          | -    | 8°   |
| Mold Draft Angle         | Ø1    | 5°          | -    | 15°  |
| Lead Thickness           | c     | 0.08        | -    | 0.23 |
| Lead Width               | b     | 0.15        | -    | 0.33 |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensions D and E1 do not include mold flash or protrusions. Mold flash or protrusions shall not exceed 0.15mm per side.
- Dimensioning and tolerancing per ASME Y14.5M.  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

## 10-Lead Plastic Micro Small Outline Package (MS) - 3x3 mm Body [MSOP]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



### RECOMMENDED LAND PATTERN

| Units                      |    | MILLIMETERS |      |      |
|----------------------------|----|-------------|------|------|
| Dimension Limits           |    | MIN         | NOM  | MAX  |
| Contact Pitch              | E  | 0.50 BSC    |      |      |
| Contact Pad Spacing        | C  |             | 4.40 |      |
| Overall Width              | Z  |             |      | 5.80 |
| Contact Pad Width (X10)    | X1 |             |      | 0.30 |
| Contact Pad Length (X10)   | Y1 |             |      | 1.40 |
| Distance Between Pads (X5) | G1 | 3.00        |      |      |
| Distance Between Pads (X8) | G  | 0.20        |      |      |

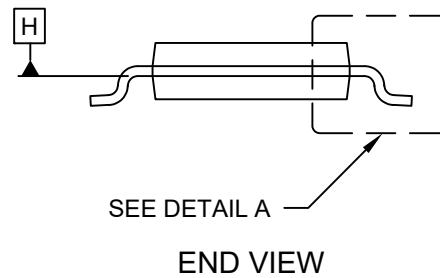
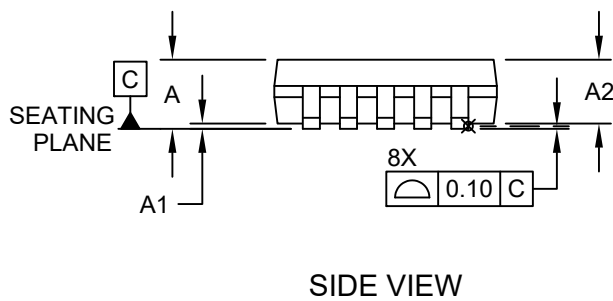
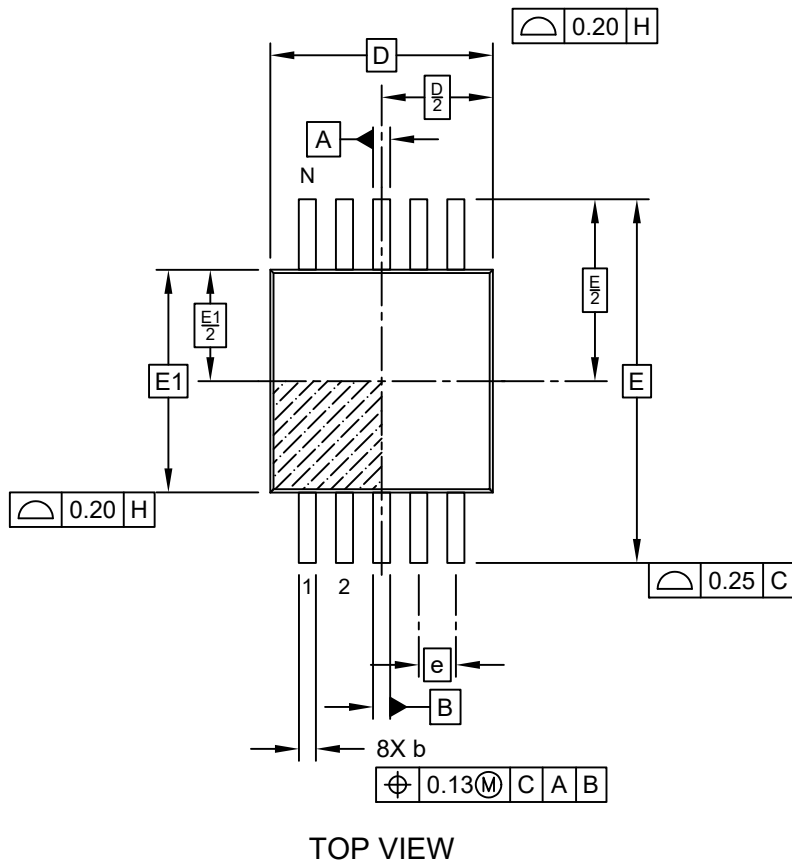
**Notes:**

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

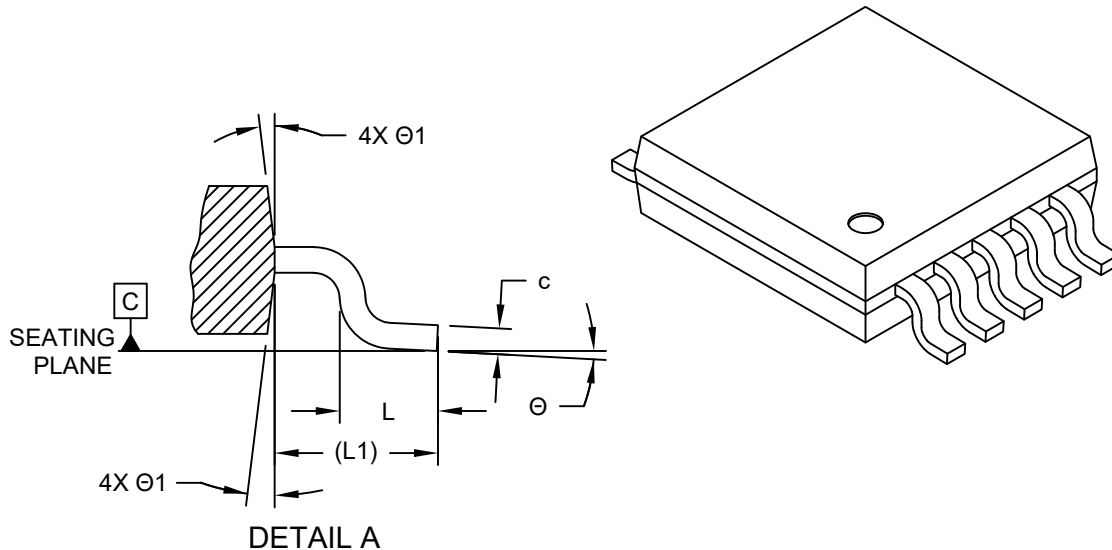
## 10-Lead Plastic Micro Small Outline Package (UN) - 3x3 mm Body [MSOP]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 10-Lead Plastic Micro Small Outline Package (UN) - 3x3 mm Body [MSOP]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



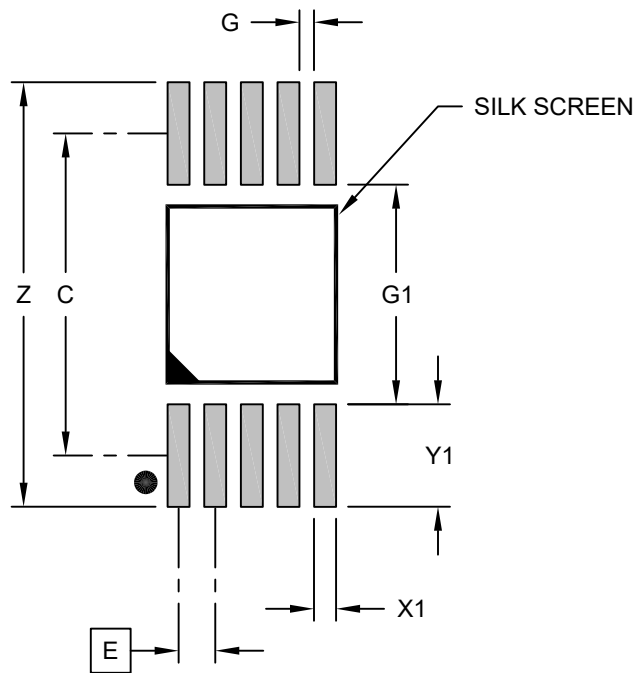
| Dimension Limits         | Units | MILLIMETERS |      |      |
|--------------------------|-------|-------------|------|------|
|                          |       | MIN         | NOM  | MAX  |
| Number of Pins           | N     | 10          |      |      |
| Pitch                    | e     | 0.50 BSC    |      |      |
| Overall Height           | A     | -           | -    | 1.10 |
| Molded Package Thickness | A2    | 0.75        | 0.85 | 0.95 |
| Standoff                 | A1    | 0.00        | -    | 0.15 |
| Overall Width            | E     | 4.90 BSC    |      |      |
| Molded Package Width     | E1    | 3.00 BSC    |      |      |
| Overall Length           | D     | 3.00 BSC    |      |      |
| Foot Length              | L     | 0.40        | 0.60 | 0.80 |
| Footprint                | L1    | 0.95 REF    |      |      |
| Foot Angle               | Ø     | 0°          | -    | 8°   |
| Mold Draft Angle         | Ø1    | 5°          | -    | 15°  |
| Lead Thickness           | c     | 0.08        | -    | 0.23 |
| Lead Width               | b     | 0.15        | -    | 0.33 |

### Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensions D and E1 do not include mold flash or protrusions. Mold flash or protrusions shall not exceed 0.15mm per side.
- Dimensioning and tolerancing per ASME Y14.5M.  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

## 10-Lead Plastic Micro Small Outline Package (UN) - 3x3 mm Body [MSOP]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



### RECOMMENDED LAND PATTERN

| Units                      |    | MILLIMETERS |      |      |
|----------------------------|----|-------------|------|------|
| Dimension Limits           |    | MIN         | NOM  | MAX  |
| Contact Pitch              | E  | 0.50 BSC    |      |      |
| Contact Pad Spacing        | C  |             | 4.40 |      |
| Overall Width              | Z  |             |      | 5.80 |
| Contact Pad Width (X10)    | X1 |             |      | 0.30 |
| Contact Pad Length (X10)   | Y1 |             |      | 1.40 |
| Distance Between Pads (X5) | G1 | 3.00        |      |      |
| Distance Between Pads (X8) | G  | 0.20        |      |      |

#### Notes:

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.